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SEMICONDUCTOR DEVICE

Abstract

A semiconductor device according to the present embodiment includes a first insulating substrate having a first surface and a second surface opposite the first surface. First and second electrically conductive layers are provided on a side of the first surface. A plurality of semiconductor chips each have a third surface facing the first surface, a fourth surface opposite the third surface, a first electrode provided on the third surface, and a second electrode provided on the fourth surface. The first electrode is electrically connected to the first electrically conductive layer. A common electrode plate has a fifth surface facing the fourth surface, is electrically connected to the second electrodes of the semiconductor chips in common, and is electrically connected to the second electrically conductive layer. A second insulating substrate is provided on a side of the second surface of the first insulating substrate.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2023-125126, filed on Jul. 31, 2023, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The embodiments of the present invention relate to a semiconductor device.

BACKGROUND

[0003] There is a case that a power semiconductor device used in an in-vehicle main inverter or the like is accommodated in a case. Meanwhile, as the configuration of power semiconductor devices becomes complicated, devices encapsulated with mold resin are increasing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 is a cross-sectional view illustrating a configuration example of a semiconductor device;

[0005] FIG. 2 is a schematic plan view illustrating a configuration example of a semiconductor module;

[0006] FIG. 3 is a schematic plan view illustrating a configuration example of the semiconductor module;

[0007] FIG. 4 is a schematic cross-sectional view illustrating a configuration example of the semiconductor module;

[0008] FIG. 5 is a cross-sectional view illustrating a configuration example of a connecting part between a semiconductor chip and a common electrode plate;

[0009] FIG. 6A is a cross-sectional view illustrating an example of a manufacturing method of a semiconductor device according to a first embodiment;

[0010] FIG. 6B is a plan view illustrating an example of the manufacturing method of the semiconductor device according to the first embodiment;

[0011] FIG. 7A is a cross-sectional view illustrating an example of the manufacturing method of the semiconductor device in continuation from FIG. 6A;

[0012] FIG. 7B is a plan view illustrating an example of the manufacturing method of the semiconductor device in continuation from FIG. 6B;

[0013] FIG. 8A is a cross-sectional view illustrating an example of the manufacturing method of the semiconductor device in continuation from FIG. 7A;

[0014] FIG. 8B is a plan view illustrating an example of the manufacturing method of the semiconductor device in continuation from FIG. 7B;

[0015] FIG. **9** is a cross-sectional view illustrating an example of the manufacturing method of the semiconductor device in continuation from FIG. **8A**;

[0016] FIG. **10** is a cross-sectional view illustrating an example of the manufacturing method of the semiconductor device in continuation from FIG. **9**;

[0017] FIG. **11** is a cross-sectional view illustrating an example of the manufacturing method of the semiconductor device in continuation from FIG. **10**;

[0018] FIG. **12** is a cross-sectional view illustrating a configuration example of a connecting part between a semiconductor chip and a common electrode plate according to a second embodiment;

[0019] FIG. **13** is a cross-sectional view illustrating a configuration example of a connecting part between a semiconductor chip and a common electrode plate according to a third embodiment;

[0020] FIG. **14** is a cross-sectional view illustrating a configuration example of a connecting part between a common electrode plate and a wire on a substrate according to a fourth embodiment;

[0021] FIG. **15** is a cross-sectional view illustrating a configuration example of a semiconductor module according to a fifth embodiment;

[0022] FIG. **16A** is a cross-sectional view illustrating an example of a manufacturing method of a semiconductor device according to the fifth embodiment;

[0023] FIG. **16B** is a plan view illustrating an example of the manufacturing method of the semiconductor device according to the fifth embodiment;

[0024] FIG. **17A** is a cross-sectional view illustrating an example of the manufacturing method of the semiconductor device in continuation from FIG. **16A**;

[0025] FIG. **17B** is a plan view illustrating an example of the manufacturing method of the semiconductor device in continuation from FIG. **16B**;

[0026] FIG. **18A** is a cross-sectional view illustrating an example of the manufacturing method of the semiconductor device in continuation from FIG. **17A**;

[0027] FIG. **18B** is a plan view illustrating an example of the manufacturing method of the semiconductor device in continuation from FIG. **17B**;

[0028] FIG. **19** is a cross-sectional view illustrating an example of a manufacturing method of a semiconductor device according to a sixth embodiment; and

[0029] FIG. **20** is a plan view illustrating a configuration example of a semiconductor module according to a seventh embodiment.

DETAILED DESCRIPTION

[0030] Embodiments will now be explained with reference to the accompanying drawings. The present invention is not limited to the embodiments. In the present specification and the drawings, elements identical to those described in the foregoing drawings are denoted by like reference characters and detailed explanations thereof are omitted as appropriate.

[0031] A semiconductor device according to the present embodiment includes a first insulating substrate having a first surface and a second surface opposite the first surface. First and second electrically conductive layers are provided on a side of the first surface. A plurality of semiconductor chips each have a third surface facing the first surface, a fourth surface opposite the third surface, a first electrode provided on the third surface, and a second electrode provided on the fourth surface. The first electrode is electrically connected to the first electrically conductive layer. A common electrode plate has a fifth surface facing the fourth surface, is electrically connected to the second electrodes of the semiconductor chips in common, and is electrically connected to the second electrically conductive layer. A second insulating substrate is provided on a side of the second surface of the first insulating substrate.

First Embodiment

[0032] FIG. **1** is a cross-sectional view illustrating a configuration example of a semiconductor device. A semiconductor device **1** includes a semiconductor module **10**, a base **20**, an adhesive **30**, a housing **40**, a fastener **50**, an encapsulation material **60**, a cover **70**, a wire **80**, and a heat-dissipation fin **90**.

[0033] The semiconductor device **1** may be a power semiconductor device that is used in an in-vehicle main inverter and switches a relatively large current with low on-resistance, for example. A stacking direction of the semiconductor module **10** with respect to the base **20** is defined as a Z-direction, one direction substantially perpendicular to the Z-direction is defined as an X-direction, and a direction substantially perpendicular to the X-direction and the Z-direction is defined as a Y-direction.

[0034] The semiconductor module **10** includes a semiconductor chip **11** and is arranged on the base **20**. The semiconductor module **10** may be a power semiconductor element such as an IGBT (Insulated Gate Bipolar

Transistor), an HEMT (High Electron Mobility Transistor), a MOSFET (Metal Oxide Semiconductor Field Effect Transistor), and a diode. The semiconductor module **10** is a high-voltage power semiconductor element with a breakdown voltage of 800 V or more, for example, and may be an SiC-MOSFET module with a rated current exceeding 400 A. The detailed configuration of the semiconductor module **10** will be described later.

[0036] The base **20** has the semiconductor module **10** mounted on its front surface. The back surface of the base **20** is made to adhere to the heat-dissipation fin **90** via the adhesive **30**. The base **20** is made of a material having high thermal conductivity, for example, AlSiC obtained by coating an SiC substrate with aluminum. The adhesive **30** is made of an adhesive having high thermal conductivity such as solder.

[0037] The heat-dissipation fin **90** has the base **20** mounted on its front surface. The back surface of the heat-dissipation fin **90** is formed in a fin shape in order to improve the heat-dissipation property. The heat-dissipation fin **90** is made of a material having high thermal conductivity such as ceramic and metal.

[0038] The housing **40** is provided on the base **20** to surround the semiconductor module **10**. The housing **40** accommodates the base **20** and the plural semiconductor modules **10** in a space surrounded by the base **20** and the housing **40**. The housing **40** is made of an insulating material such as resin.

[0039] The fastener **50** penetrates through an opening formed in the housing **40** and the base **20** and is fitted to the heat-dissipation fin **90**, **10** thereby fixing the housing **40** to the base **20** and further fixing the housing **40** and the base **20** to the heat-dissipation fin **90**.

[0040] The encapsulation material **60** encapsulates the semiconductor modules **10** in the space surrounded by the housing **40**, the base **20**, and **15** the cover **70**. The encapsulation material **60** may be a gel-like material that encapsulates a plurality of substrates **100** (see FIG. 4), the semiconductor chips **11**, a plurality of common electrode plates **120**, and the like in the above space. The encapsulation material **60** is made of an insulating material such as silicone resin and mold resin.

[0041] The cover **70** closes the opening in the housing **40** and protects the encapsulation material **60** encapsulating the semiconductor module **10**. Similarly to the housing **40**, the cover **70** is made of an insulating material such as resin.

[0042] The wire **80** is electrically connected to a portion of a wire of the semiconductor module **10** and passes through the encapsulation material **60** and the housing **40** to be led out to outside of the housing **40**. The wire **80** is made of an electrically conductive metal such as copper.

[0043] FIGS. 2 and 3 are schematic plan views illustrating a configuration example of a semiconductor module. FIG. 4 is a schematic cross-sectional view illustrating a configuration example of the semiconductor module. In FIG. 2, illustrations of common electrode plates **120g**, **120s**, **120p**, and **120n** illustrated in FIG. 3 are omitted. In FIG. 2, an electrical connection relation between each electrode of the semiconductor chip **11** and each wire on the substrate **100** is illustrated with a broken line.

[0044] As illustrated in FIG. 4, the substrate **100** as a first insulating substrate has a first surface F1 and a second surface F2 opposite the first surface F1. The substrate **100** is made of an insulating material having high thermal conductivity such as ceramic.

[0045] A wire **110** is provided on the first surface **F1** of the substrate **100**. The wire **110** as first and second electrically conductive layers is formed in any planar pattern on the first surface **F1**. The wire **110** is made of a material having high electrical conductivity and thermal conductivity, such as copper. The base **20** as a second insulating substrate is provided on the second surface **F2** side of the substrate **100**. A wire **22** on the substrate **100** is joined to the base **20** with a joining material **21**. That is, the joining material **21** is provided between the substrate **100** and the base **20** and joins them to each other. The joining material **21** is made of an electrically conductive joining material such as solder. The wire **22** is made of the same material as the wire **110**, for example, a material having high electrical conductivity and thermal conductivity such as copper. However, the joining material **21** and the wire **22** are used for thermal conduction and it is also permissible that the joining material **21** and the wire **22** are not used for electrical connection. By providing the joining material **21** and the wire **22** on the second surface **F2** side, a linear expansion coefficient is balanced between the first surface **F1** side and the second surface **F2** side of the substrate **100**, so that warpage of the substrate **100** can be prevented.

[0046] As illustrated in FIG. 2, the wire **110** on the first surface **F1** of the substrate **100** includes an anode input terminal wire **WP**, a cathode input terminal wire **WN**, a source wire **WS**, a gate wire **WG**, and an output terminal wire **WAC**. A high-voltage side waveform is formed between the anode input terminal wire **WP** and the output terminal wire **WAC**. A low-voltage side waveform is formed between the output terminal wire **WAC** and the cathode input terminal wire **WN**.

[0047] As illustrated in FIG. 2, the semiconductor chip **11** is arranged on the anode input terminal wire **WP** or the output terminal wire **WAC**. A drain (or a collector) as a first electrode is provided on a bottom surface **F3** of the semiconductor chip **11**. The bottom surface **F3** of the semiconductor chip **11** faces the first surface **F1** of the substrate **100**, and the drain (or the collector) is electrically connected to the anode input terminal wire **WP** or the output terminal wire **WAC**. The semiconductor chip **11** is joined to the substrate **100** with an electrically conductive sintered material, for example, a stacked film of **Ti**, **Ni**, and **Au**, an **Ag** film, or a stacked film of **Au** and **Ag**.

[0048] A gate electrode **G**, a source electrode **S** (or an emitter electrode), and an electrode **M** are provided on a front surface **F4** of the semiconductor chip **11** opposite the bottom surface **F3**.

[0049] As illustrated in FIG. 3, the common electrode plates **120g**, **120s**, **120p**, and **120n** are provided on the front surface **F4** of the semiconductor chip **11**. The common electrode plates **120g**, **120s**, **120p**, and **120n** connect the gate electrode **G** and the source electrode **S** of the semiconductor chip **11** and the gate wire **WG**, the source wire **WS**, and the cathode input terminal wire **WN** or the output terminal wire **WAC** to each other.

[0050] For example, as illustrated in FIG. 3, the common electrode plate **120g** is connected between each of the gate electrodes **G** of the semiconductor chips **11** and the gate wire **WG** in common. The common electrode plate **120s** is connected between each of the source electrodes **S** of the semiconductor chips **11** and the source wire **WS** in common. The common electrode plate **120p** is connected between the electrodes **M** of the upper three semiconductor chips **11** in FIG. 3 and the output terminal wire **WAC** in common. The common electrode plate **120n** is connected between the electrodes **M** of the lower three semiconductor chips **11** in FIG. 3 and the cathode input terminal wire **WN** in common. The electrodes **M** are respectively connected to the source electrodes **S** of the semiconductor chips **11**. As described above, the electrodes **G**, **S**, and **M** on the front surface **F4** of each semiconductor chip **11** are electrically connected to the wires **WG**, **WS**, **WAC**, and **WN** on the substrate **100** by the common electrode plates **120g**, **120s**, **120p**, and **120n**. In the following description, the common electrodes **120g**, **120s**, **120p**, and **120n** are also called common electrode plates **120** collectively, the wires **WG**, **WS**, **WAC**, and **WN** are also called the wires **110** collectively, and the electrodes **G**, **S**, and **M** are also called electrodes **G** and the like collectively.

[0051] The common electrode plates **120** are made of a material having electrical conductivity and thermal conductivity such as copper, aluminum, molybdenum, and tungsten. The thickness of each

common electrode plate **120** is 400 to 1000 μm , for example. As illustrated in FIG. 3, each common electrode plate **120** is a plate-like connector and has any planar shape in plan view as viewed from the Z-direction. As illustrated in FIG. 4, in the cross-section, each common electrode plate **120** extends in the X-Y plane from any of the electrodes G and the like on the front surface F4 of the semiconductor chip **11** and is bent toward the wire **110** on the substrate **100** to the -Z-direction. Accordingly, each common electrode plate **120** electrically connects any of the electrodes G and the like on the front surface F4 of the semiconductor chip **11** and the corresponding one of the wires **110** on the substrate **100** to each other.

[0052] FIG. 5 is a cross-sectional view illustrating a configuration example of a connecting part between a semiconductor chip and a common electrode plate. Each common electrode plate **120** has a fifth surface F5 facing the front surface F4 of the semiconductor chip **11**. One end of the common electrode plate **120** is electrically connected to any of the electrodes G and the like of the semiconductor chip **11** via a sintered material **140** and a plated metal **130** that are provided between the fifth surface F5 and the front surface F4 of the semiconductor chip **11**. Although not illustrated in FIG. 5, the other end of the common electrode plate **120** is electrically connected to any of the wires **110** on the substrate **100** via a sintered material and a plated metal (both not illustrated) that are provided between the fifth surface F5 and the wires **110** on the substrate **100**.

[0053] The plated metal **130** as a first plated metal is provided on the electrodes G and the like of the semiconductor chip **11**. The plated metal **130** contains any of copper, nickel, silver, gold, and palladium (Pd), for example, and is specifically made of a material having high electrical conductivity and thermal conductivity, such as NiAu and Cu. The thickness of the plated metal **130** is 1 to 15 μm , for example. Other plated metals can be made of the identical material to the plated metal **130**.

[0054] The sintered material **140** as a first sintered material is provided on the plated metal **130**. The sintered material **140** is made of any of copper, silver, lead, a tin-copper compound, a tin-silver compound, and a tin-nickel compound. The thickness of the sintered material **140** is 10 to 300 μm , for example. The sintered material **140** is a material in the form of paste or sheet when being supplied originally, and is sintered between the plated metal **130** and the common electrode plate **120** by heat treatment. Other sintered materials can be made of the identical material to the sintered material **140**.

[0055] As described above, the common electrode plates **120** are joined to the sintered material **140**, thereby being electrically connected to the electrodes G and the like of the semiconductor chip **11** via the plated metal **130**.

[0056] Side surfaces of the semiconductor chip **11** and side surfaces of the plated metal **130** are coated with an insulating layer **150**. The insulating layer **150** is made of an insulating material such as polyimide. The thickness of the insulating layer **150** is 5 to 10 μm , for example. In the Z-direction in which the semiconductor chip **11** and the common electrode plate **120** are stacked on the substrate **100**, a front surface (eighth surface) F8 of the insulating layer **150** is at a level higher than the front surface F4 of the semiconductor chip **11** and lower than a front surface (seventh surface) F7 of the plated metal **130**. Accordingly, the insulating layer **150** can protect the side surfaces of the semiconductor chip **11** and the side surfaces of the plated metal **130** while maintaining the property of dissipating heat from the sintered material **140** and the common electrode plate **120**.

[0057] The semiconductor device **1** according to the present embodiment includes the common electrode plates **120** that are each electrically connected to the corresponding electrodes G of the semiconductor chips **11** in common and that are electrically connected to the wires **110** on the substrate **100**. Accordingly, wire bonding is no longer necessary, and the electrodes G and the like of each semiconductor chip **11** and the wires **110** on the substrate **100** can be connected by the common electrode plates **120** with low resistance. By using the common electrode plates **120**, it is possible to stack the material **140** and the common electrode plates **120** on the semiconductor chip

11 and to pressurize and sinter the material **140** and the common electrode plate **120**. Accordingly, the common electrode plates **120** can connect the electrodes G and the like of each semiconductor chip **11** and the wires **110** on the substrate **100** to each other with low resistance and can provide the high heat-dissipation property to the semiconductor device **1**. It is also possible to make the semiconductor device **1** have a substantially uniform thickness.

[0058] Furthermore, use of the common electrode plates **120** in place of wire bonding enables the electrodes G and the like of the semiconductor chips **11** to be connected to the wires **110** in common at the same time. Therefore, the number of parts constituting the semiconductor device **1** can be reduced.

[0059] Next, a manufacturing method of the semiconductor device **1** according to the present embodiment is described.

[0060] FIGS. **6A** to **11** are cross-sectional views and plan views illustrating an example of the manufacturing method of the semiconductor device according to the first embodiment.

[0061] As illustrated in FIGS. **6A** and **6B**, a material to be sintered **160** is applied on the wire **110** on the substrate **100**. The material **160** is applied at a position at which the semiconductor chip **11** is to be arranged. The material **160** is made of any of copper, silver, lead, a tin-copper compound, a tin-silver compound, and a tin-nickel compound.

[0062] Next, as illustrated in FIGS. **7A** and **7B**, the semiconductor chip **11** is placed on the material **160**. The semiconductor chip **11** is arranged at the position of the material **160** applied on the wire **110** on the substrate **100**.

[0063] As illustrated in FIG. **7A**, the semiconductor chip **11** is then pressurized from the Z-direction by means of a pressurizer P. In this process, a shock absorber S is sandwiched between the pressurizer P and the semiconductor chip **11** to prevent the pressurizer P from damaging the semiconductor chip **11**. While the pressurizer P pressurizes the semiconductor chip **11**, the material **160** is heated to be sintered. Accordingly, the semiconductor chip **11** is joined to the wire **110** with the sintered material **160**, thereby being electrically connected to the wire **110** with low resistance.

[0064] Next, as illustrated in FIGS. **8A** and **8B**, the material to be sintered **140** is applied on the electrodes G and the like of the semiconductor chip **11** and on the wire **110** on the substrate **100**. The material **140** is applied at a position at which the common electrode plate **120** is to be arranged. The material **140** is made of any of copper, silver, lead, a tin-copper compound, a tin-silver compound, and a tin-nickel compound.

[0065] Next, as illustrated in FIGS. **9** and **3**, the common electrode plates **120** are placed on the material **140**. The common electrode plates **120** are arranged at the positions of the material **140** applied on the electrodes G and the like of the semiconductor chip **11** and on the wires **110** on the substrate **100**. For example, as illustrated in FIG. **3**, the common electrode plate **120g** is arranged on the material **140** applied on the electrode G of the semiconductor chip **11** and on the wire WG on the substrate **100**. The common electrode plate **120s** is arranged on the material **140** applied on the electrode S of the semiconductor chip **11** and on the wire WS on the substrate **100**. The common electrode plate **120p** is arranged on the material **140** applied on the electrodes M of the upper three semiconductor chips **11** in FIG. **3** and on the wire WAC on the substrate **100**. The common electrode plate **120n** is arranged on the material **140** applied on the electrodes M of the lower three semiconductor chips **11** in FIG. **3** and on the wire WN on the substrate **100**.

[0066] As illustrated in FIG. **9**, the common electrode plates **120** are then pressurized from the Z-direction by means of the pressurizer P. In this process, the shock absorber S is sandwiched between the pressurizer P and the semiconductor chips **11** to prevent the pressurizer P from damaging the common electrode plates **120** and the semiconductor chips **11**. While the pressurizer P pressurizes the common electrode plates **120**, the material **140** is heated to be sintered. Accordingly, the common electrode plates **120** are joined to the electrodes G and the like of the semiconductor chips **11** and the wires **110** on the substrate **100** with the sintered material **140**, thereby being electrically connected to the electrodes G and the like of the semiconductor chips **11**

and the wires **110** on the substrate **100** with low resistance. With this configuration, the semiconductor module **10** is formed.

[0067] Next, as illustrated in FIG. **10**, the semiconductor modules **10** are joined onto the base **20** with the solder **21**. The base **20** is made of a material having high thermal conductivity, such as AlSiC obtained by coating an SiC substrate with aluminum. Therefore, heat generated in the semiconductor modules **10** can be transferred to the base **20** via the wire **22** and the solder **21** and be transferred from the base **20** to the heat-dissipation fin **90** in FIG. **1** easily.

[0068] Next, the housing **40** is arranged to surround the semiconductor modules **10** on the base **20**. The housing **40** is fixed to the base **20** with the fastener **50**. At this time, the heat-dissipation fin **90** illustrated in FIG. **1** may also be fixed to the base **20** with the fastener **50** along with the housing **40**.

[0069] Next, as illustrated in FIG. **11**, the wire **80** is ultrasonic-joined to the wire **110** on the substrate **100** by means of an ultrasonic joining tool B. The wire **80** penetrates through the housing **40** and is led out to outside thereof. Accordingly, a user can supply a signal from outside or power to each semiconductor module **10** via the wire **80**.

[0070] Thereafter, the gel-like encapsulation material **60** is put on the semiconductor modules **10** in the housing **40**, and the cover **70** is fitted to the housing **40**. In this way, the semiconductor device **1** illustrated in FIG. **1** is completed.

[0071] According to the present embodiment, it is possible to stack the material to be sintered **140** and the common electrode plates **120** on the semiconductor chip **11** and to pressurize and sinter the material **140** and the common electrode plates **120**. Accordingly, the common electrode plates **120** can connect the electrodes G and the like of each semiconductor chip **11** and the wires **110** on the substrate **100** to each other with low resistance and can provide the high heat-dissipation property to the semiconductor device **1**. It is also possible to make the semiconductor device **1** have a substantially uniform thickness.

[0072] Furthermore, use of the common electrode plates **120** in place of wire bonding enables the electrodes G and the like of the semiconductor chips **11** to be connected to the wires **110** at the same time. Therefore, the number of parts constituting the semiconductor device **1** can be reduced.

Second Embodiment

[0073] FIG. **12** is a cross-sectional view illustrating a configuration example of a connecting part between a semiconductor chip and a common electrode plate according to a second embodiment. In the second embodiment, an intermediate electrode plate **122** is provided between the semiconductor chip **11** and the common electrode plate **120**. Sintered materials **141** and **142** are provided on both surfaces in the Z-direction of the intermediate electrode plate **122**. The sintered material **141** as a second sintered material is provided on the plated metal **130**. The sintered material **142** as a third sintered material is provided between the intermediate electrode plate **122** and the common electrode plate **120**. The sintered materials **141** and **142** are made of any of copper, silver, lead, a tin-copper compound, a tin-silver compound, and a tin-nickel compound. The thickness of each of the sintered materials **141** and **142** is 10 to 100 μm .

[0074] The intermediate electrode plate **122** may be made of the same material as the common electrode plate **120**. In plan view as viewed from the Z-direction, the intermediate electrode plate **122** is formed to have such a size that it can be accommodated within each of the electrodes G and the like of the semiconductor chip **11**. The thickness of the intermediate electrode plate **122** is 100 to 300 μm .

[0075] The front surface F8 of the insulating layer **150** may be at a level higher than the front surface F7 of the plated metal **130** provided on the electrodes G and the like of the semiconductor chip **11** in the Z-direction. Since the intermediate electrode plate **122** is interposed between the common electrode plate **120** and the semiconductor chip **11**, the front surface F7 of the plated metal **130** may be lower than the front surface F8 of the insulating layer **150**.

[0076] A manufacturing method of the semiconductor device **1** according to the second

embodiment is described.

[0077] After the processes described with reference to FIGS. 6A to 7B, in place of the material to be sintered **140** in FIGS. 8A and 8B, the material to be sintered **141** in FIG. 12 is applied on the electrodes G and the like of the semiconductor chip **11** and on the wires **110** on the substrate **100**, the intermediate electrode plate **122** is placed on the material **141**, and the material to be sintered **142** is applied on the intermediate electrode plate **122**.

[0078] Next, after the processes described with reference to FIGS. 9 and 3, the common electrode plate **120** is placed at the position of the material **142**. The common electrode plate **120** is pressurized from the Z-direction by means of the pressurizer P, and the material **141** and **142** is sintered. Accordingly, the intermediate electrode plate **122** is joined to the electrodes G and the like of the semiconductor chip **11** with the material **141**, and the common electrode plate **120** is joined to the intermediate electrode plate **122** and the wires **110** on the substrate **100** with the material **142**. With this configuration, the semiconductor module **10** according to the second embodiment is formed. The rest of the configuration and the manufacturing process of the second embodiment may be identical to those of the first embodiment. Accordingly, the second embodiment can obtain effects same as those of the first embodiment.

Third Embodiment

[0079] FIG. 13 is a cross-sectional view illustrating a configuration example of a connecting part between a semiconductor chip and a common electrode plate according to a third embodiment. The third embodiment further includes plated metals **131** and **132**. The plated metal **131** is provided on the fifth surface F5 of the common electrode plate **120**. The plated metal **132** is provided on a sixth surface F6 of the common electrode plate **120** opposite the fifth surface F5. The plated metals **131** and **132** contain any of copper, nickel, silver, gold, and palladium (Pd), for example, and are specifically made of a material having high electrical conductivity and thermal conductivity such as NiAu and Cu. By providing the plated metals **131** and **132** on both the front and back surfaces of the common electrode plate **120**, surface oxidation of the common electrode plate **120** (e.g., copper) is prevented. Accordingly, the common electrode plate **120** can be joined to the sintered material **140** more firmly.

[0080] The rest of the configuration and the manufacturing process of the third embodiment may be identical to those of the first embodiment. Accordingly, the third embodiment can obtain effects same as those of the first embodiment.

Fourth Embodiment

[0081] FIG. 14 is a cross-sectional view illustrating a configuration example of a connecting part between a common electrode plate and a wire on a substrate according to a fourth embodiment. The common electrode plate **120** is joined to the wire **110** on the substrate **100** with the sintered material **140**. Instead of the sintered material **140**, a plated metal (not illustrated) may be used. The plated metal contains any of copper, nickel, silver, gold, and palladium (Pd), for example, and is specifically made of a material having high electrical conductivity and thermal conductivity such as NiAu and Cu. The common electrode plate **120** can be joined to the wire **110** with the sintered material **140** or the plated metal firmly. The plated metal may be formed by a wet process, vapor deposition, ion plating, sputtering, or the like.

[0082] The rest of the configuration and the manufacturing process of the fourth embodiment may be identical to those of the first embodiment. Accordingly, the fourth embodiment can obtain effects same as those of the first embodiment.

Fifth Embodiment

[0083] FIG. 15 is a cross-sectional view illustrating a configuration example of a semiconductor module according to a fifth embodiment. The semiconductor module **10** according to the fifth embodiment includes the substrate **100** provided on the bottom surface F3 side of the semiconductor chip **11** (the fifth surface F5 side of the common electrode plate **120**) and a substrate **200** provided on the front surface F4 side of the semiconductor chip **11** (the sixth surface F6 side of

the common electrode plate **120**). That is, the substrates **100** and **200** are provided on both the bottom surface **F3** side and the front surface **F4** side of the semiconductor chip **11**. Similarly to the substrate **100**, the substrate **200** is also made of an insulating material having high thermal conductivity, such as ceramic.

[0084] A terminal **250** is provided between the substrate **100** and the substrate **200**. The terminal **250** is electrically connected to a portion of the wire **110** when pressurized by the pressurizer **P**, and functions as a terminal led out to outside of the semiconductor module **10**. For example, the terminal **250** may be electrically connected to the wire **80** in FIG. **1** and be led out to outside of the housing **40**. The terminal **250** also functions as a spacer between the substrate **100** and the substrate **200** when pressurized by the pressurizer **P**. Therefore, the terminal **250** maintains the gap between the substrate **100** and the substrate **200** so as to prevent the substrates **100** and **200** from being bent and damaged.

[0085] The substrate **200** is joined to the common electrode plate **120** with a sintered material **240** provided on the sixth surface **F6** of the common electrode plate **120**. The sintered material **240** contains any of copper, nickel, silver, gold, and palladium (Pd), for example, and is specifically made of a material having high electrical conductivity and thermal conductivity such as NiAu and Cu.

[0086] A ninth surface **F9** of the substrate **200** faces the sixth surface **F6** of the common electrode plate **120** and the front surface **F1** of the substrate **100**. A tenth surface **F10** of the substrate **200** is an opposite surface to the ninth surface **F9**. A wire **222** is provided on the tenth surface **F10** of the substrate **200**. The wire **222** is made of a material that has high electrical conductivity and thermal conductivity, such as copper. Accordingly, heat of the semiconductor chip **11** is dissipated not only via the substrate **100** but also via the substrate **200**. That is, the semiconductor chip **11** is a so-called DSC (Double Side Cooling) semiconductor device that dissipates heat from both the bottom surface **F3** and the front surface **F4** of the semiconductor chip **11**.

[0087] The semiconductor device **1** according to the fifth embodiment can improve the heat-dissipation property of the semiconductor chip **11**. The rest of the configuration of the fifth embodiment may be identical to that of the first embodiment. Accordingly, the fifth embodiment can obtain effects same as those of the first embodiment. The fifth embodiment may be combined with any of the second to fourth embodiments. With this combination, the fifth embodiment can obtain any of effects of the second to fourth embodiments.

[0088] A manufacturing method of the semiconductor device **1** according to the fifth embodiment is described.

[0089] FIGS. **16A** to **18B** are cross-sectional views and plan views illustrating an example of the manufacturing method of the semiconductor device according to the fifth embodiment.

[0090] After the processes described with reference to FIGS. **6A** to **8B**, the material to be sintered **140** is applied on the semiconductor chip **11** and the wire **110**.

[0091] Next, as illustrated in FIGS. **16A** and **16B**, the common electrode plate **120** is arranged on the material to be sintered **140** on the semiconductor chip **11**, and the terminal **250** is arranged on the material to be sintered **140** on the wire **110**. As illustrated in FIG. **16B**, the terminals **250** are arranged on the wires **WG**, **WS**, **WP**, **WN**, and **WAC** and led out to outside of the substrate **100**.

[0092] As illustrated in FIG. **16A**, the common electrode plate **120** and the terminal **250** are then pressurized from the Z-direction by means of the pressurizer **P**. In this process, the shock absorber **S** is sandwiched between the pressurizer **P** and the semiconductor chip **11** to prevent the pressurizer **P** from damaging the common electrode plates **120** and the semiconductor chip **11**. The terminal **250** functions as a spacer, thereby preventing local application of excessive pressure to the common electrode plate **120**, the semiconductor chip **11**, and the substrate **100**. While the pressurizer **P** pressurizes the common electrode plate **120** and the terminal **250**, the material to be sintered **140** is heated to be sintered. Accordingly, the common electrode plates **120** are joined to the electrodes **G** and the like of the semiconductor chip **11** and the wires **110** on the substrate **100** with the sintered

material **140**, thereby being electrically connected to the electrodes G and the like of the semiconductor chip **11** and the wires **110** on the substrate **100** with low resistance. The terminal **250** is joined to the wire **110** with the sintered material **140**, so that the wire **110** on the substrate **100** can be led out to outside of the semiconductor device **1**.

[0093] Next, as illustrated in FIGS. **17A** and **17B**, the material to be sintered **240** is applied on the common electrode plate **120**.

[0094] Next, as illustrated in FIGS. **18A** and **18B**, the substrate **200** is arranged on the material **240** on the common electrode plate **120**. The ninth surface F9 of the substrate **200** faces the common electrode plate **120** via the material **240**. The substrate **200** is also provided on the terminal **250** to sandwich the terminal **250** between the substrate **200** and the substrate **100**.

[0095] Next, the wire **222** is formed on the tenth surface F10 of the substrate **200** located on the opposite side of the ninth surface F9.

[0096] As illustrated in FIG. **18A**, the substrate **200** is then pressurized from the Z-direction by means of the pressurizer P. In this process, the shock absorber S is sandwiched between the pressurizer P and the semiconductor chip **11** to prevent the pressurizer P from damaging the substrate **200** and the like. The terminal **250** functions as a spacer, thereby preventing local application of excessive pressure to the substrate **200**. While the pressurizer P pressurizes the substrate **200**, the material **240** is heated to be sintered. Accordingly, the substrate **200** is joined to the common electrode plate **120** with the sintered material **240**. Although the terminal **250** is sandwiched between the substrate **100** and the substrate **200** and led to outside of the semiconductor device **1**, the terminal **250** can be connected to the wire **110** securely.

[0097] Next, as illustrated in FIG. **15**, the surroundings of the substrates **100** and **200** and the semiconductor chip **11** are encapsulated by the insulating layer **150**. At this time, a portion of the terminal **250** protrudes from the insulating layer **150**. Accordingly, electrical connection from outside of the semiconductor device **1** to the semiconductor chip **11** can be achieved. The wires **22** and **222** are exposed from the insulating layer **150**. This configuration enables easy heat dissipation from the substrates **100** and **200**.

[0098] The rest of the manufacturing process of the fifth embodiment may be identical to that of the first embodiment. Accordingly, the fifth embodiment can obtain effects same as those of the first embodiment.

Sixth Embodiment

[0099] FIG. **19** is a cross-sectional view illustrating an example of a manufacturing method of the semiconductor device **1** according to a sixth embodiment. The configuration of the sixth embodiment may be identical to that of the fifth embodiment.

[0100] The manufacturing method according to the sixth embodiment is described.

[0101] After the processes described with reference to FIGS. **6A** to **8B**, the material to be sintered **140** is applied on the semiconductor chip **11** and the wire **110**.

[0102] Next, as illustrated in FIG. **19**, some of the common electrode plates **120** are arranged while being connected to each other by a frame **120F**. The terminals **250** are arranged while being connected to each other by a frame **250F**. The frame **120F** couples the common electrode plates **120** to each other, and the frame **250F** couples the terminals **250** to each other.

[0103] Subsequently, as described with reference to FIG. **16A**, the common electrode plates **120**, the frame **120F**, the terminals **250**, and the frame **250F** are pressurized from the Z-direction by means of the pressurizer P. While the pressurizer P pressurizes the common electrode plates **120**, the frame **120F**, the terminals **250**, and the frame **250F**, the material **140** is heated to be sintered. Accordingly, the common electrode plates **120** are joined to the electrodes of the semiconductor chip **11** and the wires **110** with the sintered material **140** along with the frame **120F**. The terminals **250** are joined to the wires **110** with the sintered material **140** along with the frame **250F**.

[0104] Thereafter, the frames **120F** and **250F** are cut out, and the common electrode plates **120** and the terminals **250** are left on the semiconductor chip **11** and the wires **110**. Accordingly, the same

configuration as that in FIG. 16B can be obtained.

[0105] The rest of the manufacturing process of the sixth embodiment may be identical to that of the fifth embodiment. Accordingly, the sixth embodiment can obtain effects same as those of the fifth embodiment. The frame **120F** couples the common electrode plates **120** to each other, and the frame **250F** couples the terminals **250** to each other. Therefore, the number of parts to be handled in a manufacturing process of the semiconductor device **1** can be reduced.

[0106] In the sixth embodiment, the substrate **200** and the wire **222** may be provided, as in the fifth embodiment. However, the substrate **200** and the wire **222** may be omitted in the sixth embodiment.

Seventh Embodiment

[0107] FIG. **20** is a plan view illustrating a configuration example of a semiconductor module according to a seventh embodiment. The common electrode plate **120** according to the seventh embodiment has an opening OP through which the gate electrode G is exposed. The semiconductor chip **11** has the gate electrode G on its front surface, and the opening OP is formed in a region of the gate electrode G. With this configuration, the gate electrode G is electrically isolated from the common electrode plate **120** and can be electrically connected to the wire WG or the like (see FIG. **17B**) with a metal wire WR or another metal plate. A connector CN is a connecting portion between the common electrode plate **120** and the wire **110** and is a portion at which the sintered material **140** is provided.

[0108] Use of the metal wire WR in connection to the gate electrode G increases the wiring flexibility. Meanwhile, the common electrode plate **120** is used in connection to other electrodes and wires. The rest of the configuration of the seventh embodiment may be identical to that of the first embodiment. Accordingly, the seventh embodiment can also obtain the effects of the first embodiment.

[0109] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A semiconductor device comprising: a first insulating substrate having a first surface and a second surface opposite the first surface; first and second electrically conductive layers provided on a side of the first surface; a plurality of semiconductor chips each having a third surface facing the first surface, a fourth surface opposite the third surface, a first electrode provided on the third surface, and a second electrode provided on the fourth surface, the first electrode being electrically connected to the first electrically conductive layer; a common electrode plate having a fifth surface that faces the fourth surface, electrically connected to the second electrodes of the semiconductor chips in common, and electrically connected to the second electrically conductive layer; and a second insulating substrate provided on a side of the second surface of the first insulating substrate.
2. The device of claim 1, further comprising: a first plated metal provided on the second electrodes of the semiconductor chips; and a first sintered material provided on the first plated metal, wherein the common electrode plate is electrically connected to the second electrodes of the semiconductor chips by being joined to the first sintered material.
3. The device of claim 1, further comprising a joining material provided between the first insulating substrate and the second insulating substrate.
4. The device of claim 2, further comprising an insulating layer provided on side surfaces of the

semiconductor chips and side surfaces of the first plated metal, wherein a height of the insulating layer in a first direction in which the semiconductor chip and the common electrode plate are stacked on the first insulating substrate is higher than the semiconductor chip and lower than the first plated metal.

5. The device of claim 1, further comprising: a plurality of intermediate electrode plates provided between the respective semiconductor chips and the common electrode plate; a first plated metal provided on the second electrodes of the semiconductor chips; a second sintered material provided on the first plated metal; and a third sintered material provided between the intermediate electrode plates and the common electrode plate.

6. The device of claim 5, further comprising an insulating layer provided on side surfaces of the semiconductor chips and side surfaces of the first plated metal, a height of the insulating layer in a first direction in which the semiconductor chip and the common electrode plate are stacked on the first insulating substrate being higher than the first plated metal.

7. The device of claim 2, further comprising: a second plated metal provided on the fifth surface of the common electrode plate; and a third plated metal provided on a sixth surface of the common electrode plate opposite the fifth surface.

8. The device of claim 1, wherein the common electrode plate and the second electrically conductive layer are joined to each other with solder.

9. The device of claim 1, further comprising a third insulating substrate provided above a sixth surface of the common electrode plate opposite the fifth surface.

10. The device of claim 9, further comprising an electrically conductive terminal sandwiched between the second insulating substrate and the third insulating substrate and electrically connected to the first electrically conductive layer or the second electrically conductive layer.

11. The device of claim 10, further comprising a fourth sintered material provided between the common electrode plate and the third insulating substrate.

12. The device of claim 7, wherein the first to third plated metals contain any of copper, nickel, silver, gold, and palladium.

13. The device of claim 2, wherein the first sintered material contains any of copper, silver, lead, a tin-copper compound, a tin-silver compound, and a tin-nickel compound.

14. The device of claim 1, wherein the common electrode plate has openings in regions of third electrodes provided on the fourth surfaces of the semiconductor chips, respectively, and is electrically isolated from the third electrodes.

15. The device of claim 1, further comprising: a housing fixed to the second insulating substrate and accommodating the first insulating substrate, the semiconductor chips, and the common electrode plate in a space surrounded by the second insulating substrate and the housing; and a gel encapsulating the first insulating substrate, the semiconductor chips, and the common electrode plate in the space.

16. The device of claim 1, further comprising mold resin encapsulating the first insulating substrate, the semiconductor chips, and the common electrode plate on the second insulating substrate.

17. The device of claim 1, wherein the semiconductor chips each include a high-voltage power semiconductor element with a breakdown voltage of 800 V or more.

18. The device of claim 1, wherein the semiconductor chips each include an IGBT (Insulated Gate Bipolar Transistor), an HEMT (High Electron Mobility Transistor), a MOSFET (Metal Oxide Semiconductor Field Effect Transistor), or a diode using an SiC substrate.
